

DECISION CHECKLIST

Ultimate TankThilteteYt — building the one final game from three builds

Yt01 = TankThilteteYt01_v1.4.html · Yt02 = TankThilteteYt02_011.html · Yt03 = TankThilteteYt03_v1.4.html

100 decision points · companion to ANALYSIS_FINDINGS.md and ULTIMATE_GAME_BLUEPRINT.md

How to read: each question shows how **Yt01**, **Yt02** and **Yt03** handle it — the build I'd copy from carries ****(Recommended)**** on its line. The green **My Choice** : box is what the users final decision are.

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TIER 0 — FOUNDATION

★ Q01. Base codebase — which build should be the skeleton that everything else ports into?

Yt01: best feel (10-second aim AI, all 6 phased boss fights, honest meter) but no Boss Rush mode at all, only 6 evolutions, cosmetic-only skins.

Yt02: two deep code audits folded in (all 11 fixes verified real), all 3 game modes incl. Boss Rush with locked -UI gating, 12 evolutions fully wired, error overlay, 30/60 FPS cap, additive archetype skins. **(Recommended)**

Yt03: 12 evolutions wired + the best armor system + tech tree + arrow keys; closest cousin to Yt02 (198 shared functions) so its ports are the cheapest of the three.

My Choice : Build on Yt02.

★ Q02. File strategy & version naming — all three ship as one self-contained HTML file with copy-per-version discipline, but each uses a different naming scheme. What do we do?

Yt01: single file; names like TankThilteteYt01_v1.4.html (v1.x scheme).

Yt02: single file; names like TankThilteteYt02_011.html (zero-padded _NNN scheme, explicitly bans v1.x names).

Yt03: single file; names like TankThilteteYt03_v1.4.html (generation + vX.Y hybrid).

My Choice : Keep the single-file, never-overwrite rule, and use a fresh line: TankGameAi_001.html, _002, ... so ultimate builds never mix with the three legacy lines.

★ Q03. PWA package (installable offline app) — should the final game ship the install-as-app shell?

Yt01: documents a full PWA story (manifest + sw.js + icons + index redirect) — but those files weren't uploaded, so unverified; only the SW-registration script exists in the HTML.

Yt02: no PWA story; runs from the file or any static server.

Yt03: no PWA story; notes it as a future step.

My Choice : Port Yt01's documented PWA pattern (manifest + sw.js pre-caching the game file + icons + redirect index).

★ Q04. External telemetry tag — all three contain a Cloudflare analytics `<script src>` tag, which is the only external dependency in three "100% offline" games. Keep or strip?

Yt01: has the beacon; doc claims full offline (contradicted).

Yt02: has the beacon; doc claims no CDN at runtime (contradicted).

Yt03: has the beacon; doc claims no external assets (contradicted).

My Choice : Strip it in Phase 1

Q05. On-screen crash/error overlay — should runtime errors show as an on-screen banner?

Yt01: none — errors only go to console.

Yt02: has `window.onerror` + `unhandledrejection` overlay (`#ty-error-overlay`). **(Recommended)**

Yt03: none.

My Choice : Adopt Yt02's overlay, but gate it behind a hidden/settings toggle so release players don't see it; invaluable during merge testing.

Q06. Target platform — what's the primary device?

Yt01: portrait phone first (left-half joystick, right-half fire), desktop WASD secondary.

Yt02: same.

Yt03: same, plus arrow keys drive on desktop.

My Choice : Keep portrait-phone-first + desktop secondary; take Yt03's arrow-key support (cheap port) so desktop players get both input styles.

Q07. Title & tagline — what is the final game called?

Yt01: TANKTHILTETEYT branding on home screen.

Yt02: locks "TANKTHILTETEYT" + tagline "CONQUER EVERY BIOME" (its doc marks this decision as locked).

Yt03: same title family.

My Choice : Keep TankGameAi on homescreen.

Q08. Save storage key & schema — which save slot does the final game use?

Yt01: `localStorage[ 'tank_save' ]`, JSON, schema version inside; in-memory fallback wrapper when storage is blocked.

Yt02: same key/schema (same code lineage).

Yt03: same key/schema.

My Choice : All three identical — keep the same key and wrapper, bump the schema version once in Phase 0, and add a tiny migrator so existing players' coins/skins survive.

TIER 1 — CORE CONTROLS & FEEL

★ Q09. Auto-aim brain — how should the cannon pick targets?

Yt01: a 10-second cycle — for 5 s it alternates "smart slots" (boss → most dangerous → lowest HP → closest, re-rolled every 0.85 s, 60/40 priority-vs-close) with 5 s of pure closest; 16-unit local sense when only 1–2 enemies near; 25% stickiness so the gun doesn't twitch. **(Recommended)**

Yt02: simple — always closest enemy + 25% stickiness.

Yt03: same simple version as Yt02.

My Choice : Take **Yt02:** simple — always closest enemy + 25% stickiness.

Q10. Top-speed cap — should stacked speed cards + Adrenaline be allowed to make the tank uncontrollable?

Yt01: hard-caps top speed at 260% of base, in both real movement and the speed meter. **(Recommended)**

Yt02: uncapped.

Yt03: uncapped (its own doc lists this as an open balance question).

My Choice : Adopt Yt01's `CONFIG.playerSpeedMaxMult = 2.6` (movement + meter). Make the 2.6 a tunable constant so you can loosen it later.

★ Q11. Adrenaline Rush formula — what does each stack actually give after a kill?

Yt01: +25% speed per stack, no free base haste (1 stack = $\times 1.25$); also claims +5% damage per stack — but that part exists only in the HUD meter today (bug F1).

Yt02: $\times 1.25$ at zero stacks, +25% per stack (1 stack = $\times 1.50$) — applied in real movement.

Yt03: base $\times 1.25$ plus +25% per stack (effectively same as Yt02); applied in real movement.

My Choice : Merge carefully but it lasts for one minute only. There will be countdown shown, after a minute it goes back, but can always be reused — use Yt02/Yt03's real-in-movement math, and implement Yt01's +5%/stack inside the actual shell-damage formula and mirror it in the meter (fixes F1/F8 in one go). Never show a number the pipeline doesn't apply.

Q12. Fire & damage core — fire rate, shell damage, crit, multishot, pierce, splash, siege slug.

Yt01: base fire $0.25s \div \text{stat}$, damage $22 \times \text{stat}$, crit $\times 2$, multishot +1/stack capped at 4, pierce, splash $\sim 6.5u$, every 4th volley $\times 1.7$ siege slug.

Yt02: same.

Yt03: same, plus SLAM! feedback on heavy hits (covered in Q78).

My Choice : Identical in all three — keep as-is. No merge needed.

Q13. Homing missiles — the timed missile launcher and its evolution upgrade.

Yt01: missile on a timer; stacks fire more per launch.

Yt02: same base behavior; its Missile Rain evolution = faster launches, wider blasts.

Yt03: same base; its Missile Rain = 50% faster launches fired in dual salvos (verified).

My Choice : Keep shared base; same launch time but the amount stacks, it fires multiple homing missiles according to stacked amount. But capped at 10 missiles, after 10, its blasts area and damages increases

Q14. One-hit Shield Generator — the bubble that eats one full hit and recharges.

Yt01: blocks a hit, recharges in $18s \div \text{stacks}$; shows "SHIELD READY" + vibration.

Yt02: same.

Yt03: same, and explicitly ordered BEFORE armor in the damage pipeline (shield $\rightarrow$ Bastion soak $\rightarrow$ armor pool $\rightarrow$ HP).

My Choice : Identical everywhere — keep, and make Yt03's shield-before-armor ordering the canonical pipeline. Also make the shield more visible not just a circle. Make it transparent blue electrical bubble that covers the player, its gone once used until its recharges.

Q15. Combo system — kill streaks and their coin multiplier.

Yt01: 3s decay bar, coin mult up to $\times 2.2$, boss hits keep the window alive, HUD counter + feed.

Yt02: same mechanics with a combo chip UI.

Yt03: same mechanics.

My Choice : Keep Yt01's mechanics + Yt02's compact chip UI.

TIER 2 — DEFENSE & ARMOR

★ Q16. Armor pool model — how does armor work at its core? (The biggest 3-way design split.)

Yt01: flat pool = armor stat points; absorbs all damage until 0 ("ARMOR DOWN"); refills at your HP-regen rate continuously.

Yt02: flat pool = armor stat points; refills at $0.5 + 0.25 \times \text{regen}$ per second; full-height blue overlay on the HP bar with a regen pulse.

Yt03: $\text{pool} = \text{floor}(\text{maxHp} \times \text{armor}/100)$ — scales automatically with your HP builds; recharge only starts after 3s without damage, at 10%/s; separate pool logic functions; saved in mid-run saves. **(Recommended)**

My Choice : Merge — Yt03's pool math + 3s-delay recharge as the engine ($\text{armorPoolMax}/\text{recalcArmorPool}/\text{refillArmorPool}$), Yt02's full-height blue HUD overlay for the top bar, and Yt01's pickup-refill behavior (next question). Port order: damage block → regen loop → pickups → HUD → save snapshots, as one phase (P10).

Q17. Armor refills from pickups — what repair/shield pickups do to the pool.

Yt01: repair kits, shield batteries AND airdrops restore the pool to full. **(Recommended)**

Yt02: repair/shield crates restore armor too.

Yt03: repair kits + shield batteries restore; pool also heals when you pick armor/max-HP cards.

My Choice : Merge — Yt01's full-refill pickup set + Yt03's "armor/maxHp cards also top the pool up" rule. One sentence of logic in each pickup handler.

Q18. Aegis Kit consumable ("start with charged shield") — what happens if you own no armor cards?

Yt01: starts shield up; armor stays 0 if you have no armor stat.

Yt02: smart fix (verified): grants 20 base armor pool when you have no armor, so the bar and regen actually work. **(Recommended)**

Yt03: starting shield honored unless loading a save (its own v1.3 fix).

My Choice : Take Yt02's 20-base-pool fix AND Yt03's "not overwritten on load-a-save" guard — they solve different halves of the same problem.

Q19. Bastion evolution — what happens right after your shield pops.

Yt01: broken shield still soaks 25% of the next hit for 2.8s.

Yt02: same.

Yt03: same (verified identical values).

My Choice : Identical — keep as-is.

Q20. Armor HUD set — where the player can see the pool.

Yt01: top blue bar with x/y + "ARMOR DOWN" flash + pause chip.

Yt02: full-height transparent overlay on the HP bar + regen pulse + pause cell.

Yt03: all of the above plus a 3D bar floating above the tank and armor on the death screen. **(Recommended)**

My Choice : **Yt02:** full-height transparent overlay on the HP bar + regen pulse + pause cell.

TIER 3 — LEVEL-UP CARDS

Q21. Cards per level-up — how many cards are offered?

Yt01: 16 card types, 3 offered, pick 1 (stacks).

Yt02: same.

Yt03: same, except the Gold skin deals 4 cards.

My Choice : Keep 3 as the standard

★ **Q22. Early-game card guarantee — the first picks force survivability cards into the hand. Whose version?**

Yt01: first 3 upgrade picks always force Regen + Heal-on-Kill + one other — even if you already own them.

Yt02: first 5 picks force them only if you don't already own that card (ownership check). **(Recommended)**

Yt03: levels 2–6 force them, no ownership check (can repeat; its doc says user chose that).

My Choice : first 5 picks force them only if you don't already own that card

Q23. Rerolls — extra chances at a better hand.

Yt01: 1 free per run + consumable rerolls + shop "Extra Choice" (+1 per level-up).

Yt02: same.

Yt03: same, plus a tech-tree node granting up to +3 reroll ranks every run.

My Choice : Keep shared rules.

Q24. Card pool weighting — how the 3 offers are rolled.

Yt01: uniform random from the pool after forced picks; boss vault offers amplified cards + evolutions.

Yt02: doc claims weighted-by-ownership offering; code is effectively uniform + evolution injection.

Yt03: uniform + evolution injection (same as Yt02's code).

My Choice : Keep uniform (predictable), but bias boss vaults toward evolution prerequisites as Yt02's doc describes.

Q25. ELITE boss-vault cards — should beating a boss feel like a jackpot?

Yt01: boss-vault cards are amplified $\times(1.32 + 0.012 \times \text{level})$ with "ELITE" branding. **(Recommended)**

Yt02: boss vault offers bigger-weight evolution chances but no amplification.

Yt03: standard boss vault.

My Choice : Adopt Yt01's amplification into the vault alongside Yt02/Yt03's guaranteed-evolution slot (see Q40 — they combine, not conflict).

TIER 4 — EVOLUTIONS

★ **Q26. Evolution pantheon — which set of special card-combos (and how many)?**

Yt01: 6 evolutions with bespoke recipes (e.g. Cluster = 1 Missile + 1 Splash; Prism = 1 Pierce + 1 Crit; Nanite = 1 Regen + 2 Medic; Siege = 2 Damage + 2 FireRate); behavior-only effects; boss-vault only.

Yt02: 12 evolutions, strict 2×primary + 1×secondary recipe rule, each with stat bonuses + behavior, all offered and wired. **(Recommended)**

Yt03: 12 evolutions with DIFFERENT pairings than Yt02 (Prism = 2 Pierce + 1 Crit; Missile Rain = 2 Missile + 1 Multishot; Phase Lance = 2 Pierce + 1 Damage...), all wired via one effects function.

My Choice : Yt02's 12

Q27. Evolution unlock rule — when can evolutions appear in offers?

Yt01: boss vault only.

Yt02: guaranteed in boss vault + 50% chance at normal level-ups once the recipe is owned. **(Recommended)**

Yt03: same rule as Yt02.

My Choice : Yt02/Yt03's rule with 60% chance

Q28. Evolution effect style — behavior vs flat stats.

Yt01: its six are pure behavior (bomblets, post-shield soak, prism pierce, extra heal, longer kill-haste, siege slug).

Yt02: six behavior + six flat-stat (e.g. Overkill +20% dmg +1 shell; Citadel +25 HP +8 armor +1 regen).

Yt03: same split, with its own exact stat numbers (Overkill +20%/+1, Tempest Autoloader +20% FR +12% crit, Citadel +35 HP +8 armor +1 regen, PhaseLance +2 pierce +15% dmg +8% crit, Predator +4 heal +15% spd +1 adr).

(Recommended)

My Choice : Keep all six behavior evolutions (they're the fun) and use Yt03's stat-grant numbers for the other six (documented and code-verified identical).

Q29. Evolution presentation — how unlocks are shown.

Yt01: pause list + tappable detail page (materials picked/needed, active status).

Yt02: unlock banner + sound + pause progress bars + detail screen with "Grants:" bonus lines. **(Recommended)**

Yt03: detail screen showing materials/active; unlock banner added in v1.3.

My Choice : Yt02's full celebration set (banner + sound + progress bars); it's the superset.

Q30. Visible tank customization — should picked cards/evos physically change your tank?

Yt01: yes — syncTankUpgradeParts bolts visible parts on per card.

Yt02: evolution parts only via its own sync.

Yt03: visual parts for all 12 evolutions.

My Choice : Merge — Yt01's per-card part system as the framework + Yt03's 12-evo parts list as the content. One sync function, no duplicates. But when bullet multiplier selected, the tanks barrel don't increase, it remains one.

TIER 5 — ENEMIES & DIFFICULTY

★ Q31. Enemy damage curve — how fast do enemies get deadlier per level? (Three different tunings.)

Yt01: starts 1.00× at level 1, +0.032/level, then extra +0.016 after L10 and +0.022 after L20 — gentlest.

Yt02: formula $(0.70 + 0.032 \times (L-1)) \times 1.5 \rightarrow$ starts 1.05× but the steepest early slope (its doc's "∼63% cut" is really ∼36% — audit F-note).

Yt03: starts 1.00×, +0.048/level + late ramps — steepest overall. **(Recommended)**

My Choice : Don't hardcode any of them — parameterize `CONFIG.enemyDmg = { base, slope, mid, late }` with Yt03's numbers as default and Yt01's as an "easy" preset. That IS the merge: one formula, tunable presets.

Q32. Enemy HP/speed/points ramps — the other scaling numbers.

Yt01: $hp \times (1 + 0.05(L-1) + 0.03(L-10) + 0.05(L-20))$; speed capped at +35%; points ramp per level.

Yt02: same.

Yt03: same (verified identical formulae).

My Choice : $hp \times (1 + 0.03 \text{ per level})$.

Q33. Enemy roster & unlock ladder — who appears when.

Yt01: 33 types (27 regular + 6 bosses), same unlock-by-level ladder, 14 late-game variants, telegraphed snipers, artillery lob rings, mines, healer, shielded front, kamikaze.

Yt02: same roster and ladder.

Yt03: same roster and ladder.

My Choice : Identical — keep as-is (fix only the squasher/gunnier typo names if you ever touch ids — but DON'T rename ids casually; that's how Yt02's afterburn bug happened).

Q34. Elite enemies — gold-tinted tougher spawns.

Yt01: from L15, 14% of spawns, +40% HP +15% dmg ×1.7 points (doc wrongly calls them cyan — they're gold).

Yt02: same values.

Yt03: same values (and its doc is the one that describes them correctly).

My Choice : from L13, 14% of spawns, +40% HP +15% dmg ×1.7 points; fix the doc wording (F4).

Q35. Difficulty table — Easy/Normal/Hard/Nightmare multipliers.

Yt01: dmg $\times 0.7/1.0/1.3/1.6$, fire $\times 0.75/1.0/1.25/1.5$.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is (Nightmare is an achievement gate).

Q36. Enemy density scaling — how many enemies the field holds per level.

Yt01: shared density system (levelsCfg.density, max-enemies logic, spawn-rate ramp).

Yt02: same, and Level Mode exposes density as a player setting (Light→Chaos).

Yt03: same as Yt02's exposure.

My Choice : Keep the shared system with Yt02's player-facing density config.

TIER 6 — BOSSES

Q37. Boss cadence — how often bosses come and what beating them gives.

Yt01: every 5 levels, one at a time, 12s breather after, +50% HP heal on kill, normal spawns speed up while one lives.

Yt02: same cadence (heal present in code).

Yt03: same, but only 4s breather in Boss Rush.

My Choice : every 5 levels, one at a time, 10s breather after, +50% HP heal on kill, normal spawns is halved up while boss lives + heal; Yt03's 4s breather applies only inside Boss Rush (keep both numbers, they're mode-specific).

★ Q38. Boss phase fights — which bosses get multi-phase escalation, and how rich?

Yt01: ALL SIX bespoke (60%/30% HP): Warlord wider-faster fan + dual barrage + scout pair; Colossus extra 10s triple-burst; Nova double/triple rings + teleport; Titan follow-up slams; Tempest extra blinks; Fortress counter-rotating spiral arms. **(Recommended)**

Yt02: only Titan & Tempest phase; Warlord/Colossus/Nova/Fortress are single-phase.

Yt03: a generic phase system on all six (P2: $\times 0.8$ attack interval; P3: $\times 1.25$ dmg $\times 1.15$ spd) plus the Titan/Tempest specials.

My Choice : Merge — Yt01's six bespoke fights as the content, Yt03's generic enrage as the automatic fallback for any future boss that lacks a bespoke script. Best coverage, future-proof.

Q39. First-boss (Warlord) nerf depth — how gentle is the first boss?

Yt01: halves damage + projectile speed only.

Yt02: halves move speed + damage only.

Yt03: halves all four — speed 0.225, damage 10, projectile 18, interval 6.4. **(Recommended)**

My Choice : Yt03's full nerf — the first boss should teach patterns, not delete new players.

Q40. Boss Vault contents — the post-boss reward pick.

Yt01: amplified ELITE cards (see Q25).

Yt02: guaranteed evolution in the vault.

Yt03: guaranteed evolution in the vault (same rule).

My Choice : Merge — guaranteed evolution slot (Yt02/Yt03) + the other two cards ELITE-amplified (Yt01). One if in the vault hand-builder.

Q41. Boss HP scaling — bosses getting tankier over levels.

Yt01: $\times (1 + 0.035 \times \text{level} + 0.02 \times (\text{level} - 15))$.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

Q42. Boss telegraphs — fairness cues on heavy attacks.

Yt01: aim-line warning glints, Titan slam ground ring, Nova cloak-while-charging shimmer.

Yt02: same set.

Yt03: same set.

My Choice : Identical — keep as-is.

Q43. Boss HP bar extras — top bar with name is shared; add low-HP pulse?

Yt01: top bar + name only.

Yt02: top bar + name + skull pulse on radar.

Yt03: proposes (not implemented) a red pulse when boss < 25% HP.

My Choice : Keep shared bar, and implement Yt03's proposed low-HP red pulse — it's ~5 lines and helps kill-focus.

TIER 7 — WORLD & BIOMES

Q44. Biome count & rotation — the world themes.

Yt01: same 10 biomes, change every 3 levels, morph transition, 1.5s fire-hush + 3s no-spawn, radar tint. (Doc undercounts mechanics but code is full.)

Yt02: same 10 (doc wrongly says 7 — audit F3).

Yt03: same 10 (doc correct).

My Choice : 10 biomes, change every 3 levels, morph transition, 1.5s fire-hush + 3s no-spawn, no radar tint, and a longer visual transitions 10s for full biome change, there must be no sudden terrain change, it must be gradual.

★ Q45. Biome gameplay traits — should biomes affect gameplay at all?

Yt01: full dormant trait code (roots slow ~20% near living trees, heat, resonance) but disabled — `currentBiomeTrait()` returns null.

Yt02: same dormant code, hard-disabled ("always 1" per comment).

Yt03: same dormant code, disabled; its own roadmap suggests re-enabling with a visible warning indicator.

(Recommended)

My Choice : must be visual only, no effect

Q46. Water bodies — lakes/ponds in the world.

Yt01: fully removed (`hasWater:false` everywhere).

Yt02: fully removed.

Yt03: fully removed.

My Choice : stay dry.

Q47. Destructible cover — trees/rocks that block shots.

Yt01: trees/rocks have HP, block bullets, shatter with wood/rock SFX, tie into an achievement; first-boss tip teaches hiding.

Yt02: same.

Yt03: same.

My Choice : their hp must also scale, two hits of players bullet destroy each tree or rock in any level. It must scale accordingly. For low rank enemy they take longer to destroy.

Q48. Infinite streamed world — the chunk system.

Yt01: chunk-streamed terrain (48-unit chunks, double-buffered, built across frames).

Yt02: same.

Yt03: same (its doc presents it as unique, but all three share it).

My Choice : Identical — keep as-is (it's the engine's spine; never touch during the merge).

TIER 8 — PICKUPS & ECONOMY

★ Q49. Airdrop schedule — how crates arrive over a run.

Yt01: from level 1, first at +12s, then an EASED curve that thins out as you level (12s → 20s → 28s → 36s+ windows), max 2 on field. **(Recommended)**

Yt02: gated to level 4+, first +12s, then flat 40s ± 25s, max 2.

Yt03: from level 1, first +10s, flat 25s ± 20s, max 2.

My Choice : Yt01's system wholesale — the level-tiered easing is the most designed (feast early, famine late). It also carries the separate aid-orb channel (Q50).

Q50. Aid orbs — the separate floating repair/shield/haste orbs.

Yt01: own scheduler, level-gated frequencies (common early, rare later), max 2, radar pings.

Yt02: has the same orb scheduler and frequencies.

Yt03: same as Yt01.

My Choice : Identical in practice — keep the two-channel system (crates + orbs).

Q51. Early repair bias — crates healing more often in the first levels.

Yt01: first 5 levels: 70% repair crates, then fully random. **(Recommended)**

Yt02: fully random from its L4 gate.

Yt03: same 70% rule as Yt01 (identical code).

My Choice : first 10 levels: 70% repair crates, then fully random.

Q52. Crate reward table — the possible loot kinds.

Yt01: repair / overcharge / shield / coins / card / haste (XP crate removed).

Yt02: same list PLUS a leftover xp kind (inconsistency F9).

Yt03: same as Yt01 (removed XP deliberately).

My Choice : : repair / overcharge / shield / card / haste only

Q53. Black-market crates — the purple ~1-in-4 crates.

Yt01: 26% purple chance, opens a mid-run shop overlay, gold-bordered.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

★ Q54. Coin economy — how kill coins are calculated (three different tightness levels).

Yt01: kills pay points × 0.5 × combo(≤2.2) × Scavenger, then EVERY income source passes a single 10% tax — tight, scores stay small.

Yt02: kills pay full points × Scavenger — loosest.

Yt03: points × Scavenger × difficulty coin mult × Gold-skin +50% — middle, with more levers.

My Choice : Merge — adopt Yt01's single-place 10% tax + Yt03's difficulty/Gold-skin multipliers, and expose Yt01's hidden ×0.5 as CONFIG.killPayoutScale so you can tune looseness in one constant. Document the full chain in one place (F5).

Q55. Shop price growth — how repeated purchases inflate.

Yt01: growth ~1.3–1.6 per purchase; 8 unlimited sink items.

Yt02: same table.

Yt03: same shared table + separate tech-tree prices (150–600 coins).

My Choice : growth ~x5 per purchase; 8 unlimited sink items — keep; tech-tree pricing per Yt03 if adopted (Q62).

TIER 9 — EVENTS

Q56. Enemy surges — the timed pressure waves.

Yt01: L8+, first at +70s, 15s window, next at 80+rand(30), coin bounty 60+10×level.

Yt02: same, plus a 5-second "SURGE INCOMING" pre-warning banner. **(Recommended)**

Yt03: same as Yt01 (no pre-warning).

My Choice : Shared mechanics + Yt02's pre-warning (fairness).

Q57. Fun-event unlock tiers — when Bounty/Ambush/Raid/Lull can appear.

Yt01: Bounty+Ambush from L8, Raid from L10, Lull from L15 — three-tier ramp. **(Recommended)**

Yt02: Bounty+Ambush early, Raid+Lull only at L15.

Yt03: same as Yt02.

My Choice : Yt01's three-tier ramp — spreads novelty across the run.

Q58. Lull duration — how long the truce event lasts.

Yt01: 6 seconds.

Yt02: 12 seconds.

Yt03: 12 seconds.

My Choice : Pick 8s — long enough to breathe, short enough to keep tension

Q59. Active-event HUD chips — showing the current event on screen.

Yt01: bounty pill (\$\$\$) during bounty events.

Yt02: bounty pill + AMB chip for ambush. **(Recommended)**

Yt03: same as Yt02.

My Choice : Yt02/Yt03's chip set; extend with a chip for Raid/Lull when those fire.

TIER 10 — SKINS & ARMORY

★ Q60. Skin system philosophy — the second signature choice: do hulls change stats?

Yt01: cosmetic only (its handoff explicitly insists, and code confirms — no stat system).

Yt02: ADDITIVE archetype deltas on top of base (Crimson +25 DMG +10 SPD –15 HP; Emerald +40 HP +10 armor +1 regen –8 SPD; Glacier +20 SPD +2 regen –10 HP –10 DMG; Void +15 DMG +10 crit; Gold +25/+10/+5/+5/+1) — stacks rationally with shop purchases; correctly applied at run start (post-_010, revive bug fixed). **(Recommended)**

Yt03: ABSOLUTE stat presets per hull (Standard 100s; Crimson 80 HP/115 dmg/110 spd; Emerald 140 HP/8 armor/85 spd; Glacier +20 spd +10 crit; Void starts with a shield +15% fire rate; Gold +50% coins + 4 cards) — each hull is an identity, but fights the additive shop system.

My Choice : cosmetic only

Q61. Skin shop UX — how skins are presented for purchase.

Yt01: shop grid with colors/costs, live recolor everywhere incl. home showcase.

Yt02: adds archetype badge + plain-English stat description per skin. **(Recommended)**

Yt03: hull list with full stat blocks.

My Choice : Yt02's badge + description cards; if Yt03 perks are adopted, print them in the same description line.

Q62. Permanent upgrade shops — the 15-item Armory vs Yt03's extra tech tree.

Yt01: 15 Armory items (capped stats + Second Wind + Extra Choice + 8 unlimited sinks).

Yt02: same 15.

Yt03: same 15 PLUS a separate 5-node tech tree (armor/speed/shield-start/roll/damage, 150–600 coins, maxLevel 1–5).

My Choice : Merge — keep the shared 15 as the main Armory and add Yt03's 5-node tree as a second "Workshop" tab. Don't fold them together (they answer different fantasies: numbers vs loadout perks). Remove Second Wind completely as, revive is determined by coins.

Q63. Second Wind — the once-per-run auto-revive shop item.

Yt01: revive once/run; level 2 revives at 75% HP.

Yt02: same.

Yt03: same (verified).

My Choice : Remove Second Wind completely as, revive is determined by coins.

Q64. Consumables — the 5 one-run boosts.

Yt01: Lucky Charm/Head Start/Reroll/Overcharge/Aegis — same prices & growth in all three.

Yt02: same.

Yt03: same.

My Choice : let them scale per 5 purchases with x3 prices. Resets at 6th purchases. It's a loop

Q65. Achievements — the 14 lifetime awards.

Yt01: same 14 ids, rewards, lifetime stats, new-best celebration.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

Q66. Daily challenges — the stubbed bumpDaily system (9 no-op call sites in every build).

Yt01: stub only.

Yt02: stub only; its roadmap proposes 3 rotating dailies with coin rewards.

Yt03: stub only.

My Choice : remove completely, no need daily challenges, remove all codes too.

TIER 11 — MODES & PROGRESSION GATES

★ Q67. Mode lineup — which game modes ship?

Yt01: Casual (endless, named saves) + Level Mode — NO Boss Rush anywhere in the code.

Yt02: all three — Casual, Level Mode (density/difficulty/start-level, maxCleared progress gates), Boss Rush (unlock: 5 lifetime boss kills OR reach L5; locked button with  + hint; victory screen after the 6th boss; crates/surges disabled inside). (**Recommended**)

Yt03: same trio (unlock via maxCleared≥5 or maxLevel≥5, 4s between bosses).

My Choice : Yt02's

Q68. Level Mode config — how much the player can tune a structured run.

Yt01: density + difficulty + start-level fields exist.

Yt02: same, with maxCleared gating which start-levels are unlocked.

Yt03: same gating approach.

My Choice : Keep Yt02's gated config — progression integrity.

Q69. Boss Rush rules — the gauntlet's exact feel.

Yt01: (mode absent).

Yt02: hard difficulty, zero normal enemies, crates + surges disabled, victory at 6 kills, locked-UI until earned.

(Recommended)

Yt03: same concept, 4s boss-to-boss breather, victory banner.

My Choice : Yt02's ruleset + Yt03's 4s breather (Yt02 uses the shared 12s outside Boss Rush; 4s keeps the gauntlet tense).

Q70. Best-score tracking — what "best" means per mode.

Yt01: per-mode best SCORE (casual/levels separately), new-best badge on death screen.

Yt02: same score-based tracking (its doc wrongly says "highest level" — audit F-note).

Yt03: same score-based.

My Choice : Keep score-based per-mode bests; optionally add best-level as a second display stat (cheap, non-conflicting).

Q71. Casual mid-run saves — named slots + autosave.

Yt01: named slots + autosave, and they DO persist to localStorage (contrary to Yt02's doc claim).

Yt02: same persistent behavior (doc wrong — memory-only claim false).

Yt03: same, richest snapshot (field/enemies/armor/cards/evos/biome), plus death-time auto-save update at 55% HP.

(Recommended)

My Choice : Keep shared persistence and take Yt03's snapshot richness + death-time save update. With 5 seconds breather time, no enemies or bullets during this time.

TIER 12 — DEATH, REVIVE & RUN END

Q72. Revive economy — pay-to-continue pricing.

Yt01: 300 × 4ⁿ unlimited, 50% HP, 3s invulnerability, 3s no-spawn.

Yt02: same.

Yt03: same (verified identical formula).

My Choice : Identical — keep as-is.

★ Q73. Revive safety kit — what the game does to make the revive fair.

Yt01: clears all live bullets, resets the missile timer.

Yt02: clears bullets AND missiles AND hushes enemy fire for 2s (applyReviveSafety). **(Recommended)**

Yt03: base behavior only.

My Choice : Yt02's full kit — dying and instantly eating another volley feels awful; hushes enemy fire for 5s

Q74. Death screen contents — what the end-of-run screen shows.

Yt01: run stats, build summary, shop shortcut, revive button, new-best celebration.

Yt02: adds final biome + evolution count cells.

Yt03: adds live SPD/DMG/ARM meters + armor pool line + save-note. **(Recommended)**

My Choice : All three combine cleanly — they're additive UI cells, no logic conflicts.

TIER 13 — HUD & FEEDBACK

★ Q75. Combat meter architecture — how live damage/speed/armor power is displayed.

Yt01: dedicated meter rows with numeric %, bars fill-to-number, BOOST/SLOWED status labels, 0.15s refresh, speed display capped with movement.

Yt02: inline ⚡ / ✨ pills with gold glow when boosted + overcharge ⌚ countdown on the pill.

Yt03: compact 3-lane power meter (SPD/DMG/ARM) incl. an armor lane, boost glow. **(Recommended)**

My Choice : Merge — Yt03's 3-lane layout as the frame (it's the only one with armor), Yt01's BOOST/SLOWED labels + 0.15s honest refresh + cap-aware speed number, Yt02's ⌚ countdown on the damage lane. Kill the duplicate meters; one component.

Q76. HP number readout — rounding behavior.

Yt01: floors the value (never shows 100/100 while regening). **(Recommended)**

Yt02: shared base behavior.

Yt03: rounds (its doc argues bar-match; can fake full for a beat).

My Choice : Yt01's floor — honesty beats cosmetics; both are overflow-safe.

Q77. 3D armor bar above the tank — Yt03's signature polish.

Yt01: none.

Yt02: none.

Yt03: canvas sprite above the tank mirroring the pool, hides at 0. **(Recommended)**

My Choice : none. Since its already shown in — Yt03's 3-lane layout,

Q78. SLAM! hit feedback — big-hit callouts.

Yt01: none.

Yt02: none.

Yt03: "SLAM!" text + orange sparks on non-crit hits ≥45 damage or siege shots. **(Recommended)**

My Choice : Adopt — ~6 lines, pure juice, no balance impact.

Q79. Minimap feature set — what the radar shows.

Yt01: enemy dots (variants teal, target highlighted), pulsing boss ring, pulsing supply-drop pings, biome tint.

Yt02: adds retractable size (tap 160↔60) + skull glyph on bosses. **(Recommended)**

Yt03: same shared base.

My Choice : Union — Yt01's pings + Yt02's retract/skull. (Docs' "teal = elites" mislabel corrected: teal = variant types.) without biome tint.

Q80. Damage numbers & combat popups toggles — player control over number spam.

Yt01: no toggles.

Yt02: settings-backed toggles for both. **(Recommended)**

Yt03: no toggles.

My Choice : Yt02's toggles (they're settings-integrated already).

Q81. Kill feed / score popups / enemy intro cards — micro-feedback family.

Yt01: kill feed lines, score popups, "new enemy" intro splash cards, hit-stop on bosses.

Yt02: same family.

Yt03: same family.

My Choice : Identical — keep as-is. But make it smaller more transparent and lasts shorter

Q82. Tutorial toasts — first-run guidance.

Yt01: ~13 tips (drive, fire, cards, crates, boss-cover tip) with seen-registry.

Yt02: ~12.

Yt03: ~12.

My Choice : Yt01's set (superset by one tip).

Q83. Reduced-flash mode — what "reduced" means for damage flashes.

Yt01: shortens flash 80ms → 45ms. **(Recommended)**

Yt02: removes the flash entirely.

Yt03: removes it entirely (same as Yt02).

My Choice : Yt01's shorten-don't-remove (a cue should still exist, just gentler); matches its doc too.

TIER 14 — AUDIO, HAPTICS & SETTINGS

Q84. Audio stack — fully synthesized sound.

Yt01: SFX suite (shoot/crit/explosion/shatter/hurt/heal/levelup/card/coin/boss alarm+fanfare/revive/achievement), engine hum, per-biome wind, generative music following combat intensity.

Yt02: same; documents that hit/kill are silent stubs by design.

Yt03: same.

My Choice : Identical — keep as-is.

Q85. Music & ambience defaults — out-of-box audio state.

Yt01: music toggle present, defaults sound ON.

Yt02: same.

Yt03: same.

My Choice : Identical — keep defaults.

Q86. Haptics patterns — vibration moments.

Yt01: level-up, evolution, shield-ready, blocked hit, boss down + master toggle.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

Q87. Settings panel — which toggle set is canonical.

Yt01: ~10 toggles (sound, music, graphics, camera, assist, haptics, hands, shake, flash).

Yt02: 13 toggles — adds damage numbers, combat popups, 30/60 FPS, reset-all. **(Recommended)**

Yt03: same shared core.

My Choice : Yt02's 13 — the superset; all persist already.

Q88. Camera modes — follow/wide + toggle button.

Yt01: follow/wide, smooth lerp, shake scaling in wide.

Yt02: same + floating  button (its 'normal' bug fixed and verified).

Yt03: same.

My Choice : **Yt01:** follow/wide, smooth lerp, shake scaling in wide.

Q89. Auto-quality governor — automatic graphics dropping under load.

Yt01: watches enemy/bullet/particle load, drops to Low after ~2 sustained windows, restores + notifies.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

Q90. FPS throttle — 30/60 cap setting.

Yt01: none.

Yt02: frame-interval gated 30/60 setting. **(Recommended)**

Yt03: none.

My Choice : Yt02's — battery saver for weak phones; comes with Q87.

Q91. Left-handed layout & control assist — accessibility pair.

Yt01: swap joystick/fire sides; assist speeds turret tracking.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

TIER 15 — ENGINE & PIPELINE (ADVANCED)

Q92. Render budget techniques — why it runs on mid phones.

Yt01: pixelRatio ≤ 1.25 , shared cached geometries/materials, scratch-vector reuse, render-on-demand when idle.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is; never regress these during ports.

Q93. Chunk micro-tasking — world building spread across frames.

Yt01: runChunkTasks budget per frame; biome morphs queue-drained.

Yt02: same.

Yt03: same.

My Choice : Identical — keep as-is.

Q94. Hit-stop — the 40ms freeze on boss hits.

Yt01: present.

Yt02: present.

Yt03: present.

My Choice : Identical — keep as-is.

Q95. Error containment — crash resistance.

Yt01: try/catch around the main loop (one bad frame never kills the game).

Yt02: same + the error overlay (Q05).

Yt03: same.

My Choice : Keep both layers (loop guard + overlay).

Q96. Validation discipline — how every future change gets checked.

Yt01: syntax-check all blocks; smoke-test extracted functions against stub DOM; md5 checks.

Yt02: strictest: copy-first rule, node --check, a 19-item must-have grep list, "never present a file with a failed check".

Yt03: syntax check + targeted greps + "ship an in-game test checklist with every build".

My Choice : Merge into one harness (blueprint §7): Yt02's must-have list extended with all ported ids + Yt03's in-game checklist convention + Yt01's meter-truth smoke (meter value must equal pipeline value).

Q97. Documentation convention — keeping the handoff honest.

Yt01: two docs (status + diary); had a duplicated paragraph; otherwise accurate.

Yt02: one giant doc; several stale sections survived (revive/§9/biome count).

Yt03: one doc — the most accurate of the three.

My Choice : One living doc for the ultimate build, regenerated every release from a fixed template (audit F10) — I'll own it.

Q98. Balance constants centralization — where tuning numbers live.

Yt01: scattered but commented (its v-batch comments are helpful).

Yt02: scattered.

Yt03: scattered.

My Choice : Merge in Phase 8 — one CONFIG block owning: enemy curve, adrenaline numbers, armor rates, crate intervals, kill payout chain, speed cap. Nothing tuned ad hoc after that.

Q99. Scope guard — does the merge add NEW gameplay?

Yt01: n/a (baseline).

Yt02: n/a.

Yt03: n/a — but its roadmap lists future ideas (dailies, biome traits, boss-bar pulse, save swatches).

My Choice : v1.0-ultimate = ports + fixes ONLY. New features needs users permission, you can present the ideas.

Q100. Definition of done for v1.0-ultimate — what "finished" means.

Yt01: "feature-complete, testing/tuning/publishing remains."

Yt02: "zero known bugs, handoff-clean."

Yt03: "all scripts compile, ~50 assertion checks passing, known limitations listed honestly."

My Choice : Merge the strictest bar: all ★ questions resolved + Phases 0–9 green + must-have list green + 30-minute play test across all 3 modes and all 6 bosses + save round-trip + zero console errors with the overlay on.

SECTION T — TIER A: HOME SCREEN & FRONT DOOR

★ Q101. Home screen layout — what greets the player and which buttons exist?

Yt01: 3D hangar with the tank on a display platform wearing your equipped skin; three stat cells (coins, best Casual, best Level Mode); buttons: Casual, Level Mode, Shop, Awards, Settings — five buttons, no Boss Rush.

Yt02: same hangar + sixth button Boss Rush, always visible but greyed with  and a hint line until earned.

(Recommended)

Yt03: same as Yt02 (six buttons incl. Boss Rush with its own unlock gate).

My Choice : Keep the 7-button hangar.

Q102. "What you're driving" line — a small label on the home screen naming your equipped hull.

Yt01: HAS it — `updateHomeSpec()` writes your skin's name into a home-spec element under the buttons.

(Recommended)

Yt02: no such line.

Yt03: no such line.

My Choice : Adopt Yt01's little label — a few lines of DOM text, and with stat-carrying skins (v1 Q60) telling players which hull is equipped becomes genuinely important. Extend it with the hull's one-line stat description from Yt02's shop badges.

Q103. Drag-to-spin showcase tank — spinning the tank on the home screen.

Yt01: touch-drag spins the showroom tank; it wears your equipped skin and upgrade parts.

Yt02: same.

Yt03: same.

My Choice : Identical everywhere — keep as-is.

Q104. The "erase everything" button — should players be able to wipe all progress from inside the game?

Yt01: does not exist — no reset anywhere in its code.

Yt02: "Reset All Data" button with a confirmation dialog. **(Recommended)**

Yt03: "⚠️ Reset All Progress" button with a confirmation dialog.

My Choice : Keep the button (Yt02 or Yt03 flavor — functionally identical double-confirm). Players on shared devices inevitably need a fresh start, and a confirm dialog is enough protection. Place it last in Settings in red, as both builds do.

SECTION T — TIER B: CAMERA & SCREEN FEEL

Q105. Cinematic camera behavior — the automatic zooms and leans during play.

Yt01: dynamic rig — camera distance responds to the battlefield (documented as zoom-in when enemies crowd you, zoom-out when clear, extra during bosses) and leans into your driving direction at speed.

Yt02: same shared rig.

Yt03: same shared rig.

My Choice : Identical family — keep as-is. (The exact zoom triggers are shared camera code; no decision to make beyond keeping it.)

✕ **Q106. Screen-shake settings — how many levels should the shake toggle have?**

Yt01: THREE states — Full / Reduced / Off (a real 3-way cycle in code). **(Recommended)**

Yt02: TWO states — binary toggle labelled Low/Full (its handbook's "Full/Reduced/Off" table overstates the code).

Yt03: TWO states — binary toggle labelled Off/Full.

My Choice : Yt01's three states — motion-sensitive players need a true Off, and Reduced keeps a cue for everyone else. Also corrects both handbooks' claims (this is the code truth: only Yt01 implements 3 states).

SECTION T — TIER C: FEEDBACK & DIAGNOSTICS

✕ **Q107. Haptics richness — how many distinct vibration patterns exist.**

Yt01: 11 vibration call sites — distinct patterns for level-up, evolution, shield-ready, blocked hit, boss down, armor break, pickups, and more. **(Recommended)**

Yt02: 5 call sites.

Yt03: 5 call sites.

My Choice : Yt01's fuller set — vibration is free feedback on phones. If any pattern feels noisy in playtesting, trim then; starting from rich is easier than adding later.

✕ **Q108. Built-in diagnostics — crash banner and hidden debug readout.**

Yt01: neither — errors vanish into the console.

Yt02: BOTH — a red on-screen error banner (window.onerror overlay) AND a hidden diagnostic readout (tap the HUD 3x: tapForDiag/diagTick) with live FPS and object counts. **(Recommended)**

Yt03: the hidden diagnostic readout only (no error banner).

My Choice : Adopt both from Yt02, gated behind a debug flag so normal players never see them (refines v1 Q05). Bonus correction: the diag readout is NOT Yt02-exclusive — Yt03 has it too; the error banner is the Yt02-only piece.

SECTION T — TIER D: REVIVE & ECONOMY

★ ✕ Q109. Revive safety package — what happens the moment you pay to come back. (CORRECTED — this flips v1 Q73.)

Yt01: the full package, actually wired inline: 50% HP, 3 s invulnerability, ALL enemy bullets deleted, missile timer reset, 3 s spawn block, effects cleaned up. **(Recommended)**

Yt02: 50% HP + 3 s invulnerability ONLY — its `applyReviveSafety()` helper (bullets/missiles/hush) exists but is never called by anything.

Yt03: 50% HP + 3 s invulnerability only — you wake up inside the bullet storm that killed you.

My Choice : Yt01's package — the re-audit proved it's the ONLY build whose safety steps actually execute (Yt02's helper is dead code, Yt03 has none). Port Yt01's inline steps into the shared `buyContinue`. Optionally also wire Yt02's never-called helper as the tidy function and CALL it — same effect, cleaner code.

✕ Q110. Aid-orb repair odds — how often the floating orbs heal (the new set claimed all three identical; they are not).

Yt01: two tiers — levels ≤7: 45% repair / 30% shield / 25% haste; after: 34/33/33.

Yt02: THREE tiers — levels ≤5: 70% repair; levels 6–7: 45%; after: 34%. Smoothest difficulty descent. **(Recommended)**

Yt03: two tiers — levels ≤5: 70% repair / 15% shield / 15% haste; after: even thirds.

My Choice : Yt02's three-tier table — generous exactly when runs are fragile (first 5 levels), then easing off in steps. It's a drop-in constant table either way.

Q111. Shopping from the death screen — do purchases made while dead apply to the run you're about to revive into?

Yt01: yes — revive re-applies the meta delta so death-screen buys affect the continued run.

Yt02: yes — same `applyMetaDelta` call in its revive.

Yt03: yes — same mechanism (verified in its `buyContinue`).

My Choice : Identical everywhere — keep as-is.

✕ Q112. The difficulty coin multiplier slot — should harder difficulties pay more coins?

Yt01: no such concept — difficulty changes danger only.

Yt02: no such concept.

Yt03: has a code slot for it — its coin formula reads `difficulty.coins`, but the difficulty table never defines a coins value, so the slot is inert (always ×1). **(Recommended)**

My Choice : Wire Yt03's dormant slot ON deliberately: Easy ×1.0 / Normal ×1.1 / Hard ×1.25 / Nightmare ×1.5 — tougher runs should pay more, and the pipe already exists (one table edit). If you prefer flat income, delete the dead slot instead — but decide, don't leave it half-wired.

✕ Q113. Free reroll baseline — does every run start with a free reroll? (CORRECTED — this fixes v1 Q23.)

Yt01: ZERO free rerolls — budget starts empty; rerolls come from the consumable and shop "Extra Choice" only.

Yt02: ZERO free rerolls — same as Yt01.

Yt03: 1 FREE reroll every run, plus tech-tree ranks (+1 each) and Extra Choice on top. **(Recommended)**

My Choice : Yt03's 1-free baseline — removes "dead hand" frustration at zero balance cost, and stacks cleanly with the paid sources. (This answer supersedes v1 Q23's incorrect "all three have 1 free".)

Q114. Overkill Array's exact bonus — one evolution's numbers differ between the two 12-evo builds.

Yt01: evolution doesn't exist in its 6-evo pantheon.

Yt02: +20% damage, +1 shell, +1 **splash** (explosive impact on top).

Yt03: +20% damage, +1 shell — no splash term.

My Choice : Yt03's documented numbers as the base (v1 Q28) PLUS Yt02's +1 splash on Overkill specifically — splash synergizes with the extra shell and makes the evolution feel appropriately "array"-sized. One number in one bonuses object.

SECTION T — TIER E: SAVES & DATA INTEGRITY

★ Q115. Evolution progress surviving a resume — Yt03's unfixed cousin of a bug Yt02 already solved.

Yt01: counts live in `runCardStats`, which its snapshot preserves.

Yt02: fixed in `_010` — `runCardsObj` was added to `snapshotRun()` after the same bug was found there.

(Recommended)

Yt03: BUGGED — `runCardCounts` is absent from its snapshot and resume, so after loading a mid-run save all evolution recipes read 0 and can never unlock.

My Choice : Fix Yt03's bug the Yt02 way — add `runCardCounts` to `snapshotRun()` and restore it on resume. Two lines, prevents a run-ruining surprise. Mandatory regardless of other choices.

★ Q116. Tech tree persistence — adopt Yt03's tech tree WITH its save bug fixed (extends v1 Q62).

Yt01: no tech tree.

Yt02: no tech tree.

Yt03: five research lines (armor ×5, speed ×5, starting shield ×1, reroll ×3, damage ×5) — but purchases write to `state.tech`, which `saveGame()` never includes, so all research vanishes on page close. **(Recommended)**

My Choice : Adopt the tree (per v1 Q62) and add `tech: state.tech` to the save payload + restore in `loadGame()`. While touching it, re-check prices so the tree complements rather than duplicates the 15-item Armory (armor/speed/damage ranks are small and cheap vs the Armory's big steps — that division works).

★ Q117. Save validation on load — protecting the game from damaged or outdated saves.

Yt01: loads with no safety checks — a corrupted/old save can throw at startup.

Yt02: same (loads blind).

Yt03: same.

My Choice : Add a load-time validator to the master build: version tag check, try/catch around parse, per-field type guards with defaults, and a one-shot migrator when the schema bumps (v1 Q08). A bad save should ever only cost its own data, never a crash.

Q118. Skid-mark policy — the tread marks your tank leaves.

Yt01: hard-capped at 36 meshes, each living 3.4 s — tightest memory budget.

Yt02: same 36-cap, 3.4 s life.

Yt03: NO count cap — marks live 10 s each and are culled only by age, so roughly 3× more decals exist at peak (the new set's "uncapped" observation, confirmed). **(Recommended)**

My Choice : Take Yt03's longer, nicer trail but add Yt01/Yt02's hard 36-mesh cap on top — best look, bounded memory on weak phones.

SECTION T — TIER F: HYGIENE & RELEASE

Q119. Dead-code cleanup — remove the catalogued dormant leftovers in the master build?

Yt01: carries dormant trait bodies (biome roots/heat/resonance, unreachable because of the null gate) and a duplicated doc paragraph.

Yt02: carries the never-called `applyReviveSafety` (unless Q109 wires it), the unreachable SLOWED pill state, plus stale handbook sections.

Yt03: carries the inert difficulty-coin slot (unless Q112 wires it), dormant trait bodies, uncapped skid meshes (unless Q118 caps them), and honest-but-dead trait code.

My Choice : Full cleanup pass at the end of the merge — every "dead" item either gets its living replacement wired (per the answers above) or is deleted. The master build should contain zero known-dead code, matching Yt02's "handoff-clean" ideal.

★ **Q120. Master build naming — what the final game's files are called. (Supersedes the naming half of v1 Q02.)**

Yt01: TankThilteteYt01_v1.4.html style (generation_version).

Yt02: TankThilteteYt02_011.html style (zero-padded build number).

Yt03: TankThilteteYt03_v1.4.html style (generation_version). **(Recommended)**

My Choice : already answered

Q121. Lifetime records — the forever-stats and celebrations.

Yt01: lifetime kills, distance, playtime, coins earned, max level, max combo + "new personal best" celebration on the death screen.

Yt02: same.

Yt03: same.

My Choice : Identical everywhere — keep as-is.

Q122. Level-up queue & boost pills — two small shared behaviors, confirmed.

Yt01: leveling twice mid-card-pick queues the choices one at a time (pendingChoices); timed boosts show as small status pills (shield/overcharge/haste/bounty).

Yt02: same queue (7 refs) + same pills + event chips.

Yt03: same queue + same pills.

My Choice : Identical everywhere — keep as-is.

★ **Q123. Source structure — develop the master build as one hand-edited file (as all three do), or split the source into readable files with a build step that glues them into the shipped single file?**

Yt01: one 1.1 MB file, hand-edited, per its own convention.

Yt02: one file; its 7,400-line script is exactly where three duplicate functions and a dead counter survived eleven patches.

Yt03: one file, same discipline. **(Recommended)**

My Choice : Merge — keep the SHIPPED artifact a single HTML file (offline, double-clickable — that stays non-negotiable), but develop from split sources (data / systems / UI) with a one-command concatenation build. Reviewable code is the only realistic way to stop the duplicate-definition bug class (C16) from recurring. This breaks all three builds' stated convention deliberately — that convention is what allowed the defects.

Q124. Version history — keep the copy-a-new-file-per-version convention, or move history to Git?

Yt01: copy-per-version, five frozen files listed.

Yt02: copy-per-version called "the hard law of this project", eleven files listed.

Yt03: copy-per-version plus a baseline checksum to detect tampering. **(Recommended)**

My Choice : keep the copy-a-new-file-per-version

★ **Q125. Old-save migration — what happens to existing players' saves when the master build loads them?**

Yt01: saves use tank_save with skins stored under id ice for Glacier.

Yt02: same key and same ice id; schema version 3.

Yt03: same key, but Glacier is stored as glacier — a Yt02 save loaded here would carry an id this build doesn't define. **(Recommended)**

My Choice : Merge — keep the same storage key (players keep coins/skins/achievements), add a one-time migrator: rename ice → glacier, fill any missing fields with defaults, validate shape (per v2 Q117), bump the schema version. Nobody loses progress; the id landmine (C29) is defused.

Q126. Locked Boss Rush button behavior — how does the locked mode look and behave?

Yt01: no Boss Rush.

Yt02: always visible, greyed to 42% opacity, unclickable,  + "Kill 5 bosses to unlock", and the lock re-evaluates LIVE (the moment you qualify it unlocks, no reload). **(Recommended)**

Yt03: visible and dimmed, but the lock is only checked once at page load (stays stale until refresh), and tapping it while locked fires a browser alert() popup — blocking and jarring on a phone.

My Choice : Yt02's version wholesale — live-updating, non-blocking, self-explanatory. No merge needed; Yt03's stale-check + alert() is strictly worse.

Q127. One clock for everything — should firing rate move to the pause-safe clock like every other timer?

Yt01: fire-rate timing uses the wall clock (keeps running during pause); boosts/armor/spawns use the pause-safe clock.

Yt02: same split.

Yt03: same split — its own developer notes name the pause-safe clock as the intended rule, but firing was never converted. **(Recommended)**

My Choice : Merge — one pause-safe clock everywhere. One-line change per build, removes a whole class of invisible timing drift, and matches the stated design intent.

Q128. Minimum damage floor — after armor soaks a hit, must at least 1 damage always leak through?

Yt01: no floor — a fully-soaked hit costs zero, exactly as the armor text promises. **(Recommended)**

Yt02: hidden floor — anything that gets past the pool costs at least 1 HP (contradicts its own "absorbs ALL damage" description).

Yt03: no floor.

My Choice : Yt01/Yt03's no-floor. A shield that always leaks contradicts its own card text and quietly buffs swarm enemies. If chip-damage creep worries you, tune enemy fire rates instead — visibly, not through a hidden constant.

Q129. Pause screen layout — how is your build presented when paused?

Yt01: core stat rows, a deduplicated card list with xcounts, evolutions with tappable detail pages.

Yt02: a 5-column stat grid (10 cells, vitals first: Max HP / Shield / Regen / Heal-kill / Crit over Speed / Damage / Fire rate / Multishot / Pierce), then cards, then evolution progress bars; the damage cell mirrors the HUD's Overcharge glow. **(Recommended)**

Yt03: a compact two-column layout plus its three-lane power meter.

My Choice : Merge — Yt02's 5-column grid as the frame (densest, best-prioritized), Yt03's meter lane embedded at the top, Yt01's tappable evolution detail pages kept.

Q130. Damage-direction indicator — an on-screen arc pointing at whoever shot you.

Yt01: no such feature.

Yt02: works — element, rotation math and fade all wired. **(Recommended)**

Yt03: the math and CSS exist but the HTML element was never added, so it no-ops on every hit (dead work 60x/sec once you're under fire).

My Choice : Adopt Yt02's working version (element + logic). It answers "where is that sniper?" — essential when damage comes from off-screen.

★ Q131. Evolution gating engine — which recipe-checking system becomes canonical?

Yt01: counts incremented and checked correctly — but only 6 evolutions with mixed recipe sizes (1+1, 2+2).

Yt02: advertised 2+1 counts are never counted; offers fall back to ownership-only (1+1 in practice); pause progress bars read a permanently-zero counter; Cluster and Missile Rain share one recipe.

Yt03: counts incremented, checked exactly as advertised, all twelve recipes distinct. **(Recommended)**

My Choice : Yt03's engine as canonical, plus one de-duplication: change Missile Rain's recipe so it no longer clones Cluster Warheads' (e.g. 2 Missile + 1 Multishot — matching its own "rain = many launchers" fantasy). Keep count-based requires{} as the single source of truth and feed BOTH the offer logic and the pause progress bars from it — one counter, incremented in one place.

Q132. ELITE vault flavor — now that we know BOTH Yt01 and Yt03 amplify boss-vault cards, which amplification?

Yt01: scaling amplification — $\approx \times 1.32$ at level 1, rising to $\approx \times 1.6$ at level ~23+, so late-game vaults stay exciting. **(Recommended)**

Yt02: no amplification.

Yt03: flat $\times 1.30$ with "★ ELITE" branding and a "+30% Elite Vault Boost" description line.

My Choice : Yt01's curve with Yt03's copywriting — the scaling keeps vaults relevant at level 30, and Yt03's clearer card text tells the player what happened. One formula, one label.

Q133. Evolution hints on offer cards — should cards tell you what they're building toward?

Yt01: yes — each offer card can show which evolutions it feeds (and "· online" once built), via its role-lookup helper. **(Recommended)**

Yt02: no card hints; progress only in the pause screen.

Yt03: no card hints.

My Choice : Yt01's hints — "this is 1 of the 2 Missile Pods you need for Cluster Warheads" is the difference between deliberate build-crafting and picking blind. Cheap, and it makes the (fixed) evolution engine legible.

Q134. Afterburner spelling standard — one id everywhere.

Yt01: data says afterburner, but its tank-part code checks afterburn — the visual never renders (the speed effect works).

Yt02: standardized on afterburn everywhere (its _010 fix).

Yt03: standardized on afterburner everywhere. **(Recommended)**

My Choice : afterburner — it's the word players actually see. Grep-enforce it: after the merge, exactly one spelling may exist in the file (add it to the must-have checks). This bug class has now bitten two of the three builds.

Q135. Card & evolution text casing.

Yt01: Title Case throughout ("Adrenaline Rush", "Cluster Warheads").

Yt02: Title Case in card lists but ALL CAPS in its evolution data table ("NANITE HARVEST").

Yt03: Title Case throughout. **(Recommended)**

My Choice : Title Case everywhere — caps in small card frames read as shouting and scan worse. One pass over the evolution data table settles it.

Q136. Documentation generation — how do the master build's docs stay truthful?

Yt01: two hand-written documents (status + diary); accurate but hand-maintained (one duplicated paragraph slipped in).

Yt02: one large hand-written handoff; several sections went stale and its integrity list text-matches strings rather than testing behavior.

Yt03: one hand-written diary — the most accurate of the three. **(Recommended)**

My Choice : Merge — Yt03's single-diary shape plus generated numbers: a small script reads the actual CONFIG/data tables and emits the stat tables into the doc. Hand-written prose, machine-generated numbers — that's the only durable cure for documentation drift (their idea; adopted).

Q137. Release validation — text-matching checklists or real behavior tests?

Yt01: syntax check + smoke-tested functions against a stub DOM.

Yt02: syntax check + a 19-string must-have list — which passes today even though the systems it names are broken (C16–C18 prove text matching can't catch behavior).

Yt03: syntax check + targeted greps. **(Recommended)**

My Choice : Merge — keep the syntax check and must-have greps as SMOKE checks, and add the four behavior tests that would have caught every PO found across all three audits: (1) coin payout with N Scavenger = expected value exactly once; (2) evolution unlock requires the stated counts; (3) fully-soaked armor hit costs 0 HP; (4) revived player has a real spawn-pause. Four assertions, run per release.

Q138. Showcase tank engineering — the home/pause 3D preview.

Yt01: home tank wears your skin AND your installed upgrade parts; separate canvas handling.

Yt02: re-parents the ONE menu canvas between the home screen and the pause screen — no second context, no duplicate allocation. **(Recommended)**

Yt03: standard showcase.

My Choice : Merge — Yt02's canvas re-use as the mechanism + Yt01's upgrade-parts display as the content. Better engineering and better storytelling on the same model.

MANDATORY FIXES — no decision required (found & scheduled)

These are correctness repairs folded into the build regardless of any answer above:

- **Deduplicate Yt02's three double-defined functions** (addKillReward, updateCombo, quickSaveFromPause) — keep exactly one of each (C16).
- **One coin function, each multiplier applied once** — kills the Scavenger/Lucky-Charm double payout (C17); fold in the v1 Q54 economy decision.
- **One evolution counter, incremented in one place, feeding offers AND pause progress** — per Q131 (C18).
- **Wire Yt01's full revive-safety into the shared revive** (v2 Q109/C20) — including making any "3s safe" message TRUE.
- **Snapshot Yt03's `runCardCounts`** (v2 Q115) and **persist the tech tree** (v2 Q116) if adopted.
- **One `afterburner` spelling** (Q134) and **add Yt03's missing `#dmg-direction` element** if Q130 adopts it.
- **Strip injected Cloudflare tags** (v1 Q04) and **migrate `ice` → `glacier`** (Q125).